Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science, Probability for Data Science, Introduction to Urban Data Analytics; *In progress:* Introduction to Machine Learning, Efficient Algorithms & Intractable Problems

Technical Skills

Languages: Python, SQL, Java

Product: User Research, User Interviews, Discovery Read-outs, Documentation, Cross-Functional Collaboration

Data & ML: pandas, NumPy, scikit-learn, XGBoost, PyTorch, PostgreSQL, ArcGIS Online, Jupyter

Engineering: FastAPI, REST APIs, AWS S3, pytest, Git, GitHub

Experience

The After Collective | Fung Fellow, Health + Innovation | Berkeley, CA **Aug 2026 - Present**

- Helped run **7** interviews with wildfire survivors and leaders (led **2**), finding insurance a shared struggle.
- Turned that into **3** AI-assisted concepts with teammates, took on a checklist tool for survivors, and gathered reactions.
- Took on testing the partner's recovery coach for gaps, and organize the team's documentation.

Guang Labs | Engineering Intern | Remote **May 2026 - Aug 2026**

- Built a service for an AI video startup that sorts videos and images into **6** content types, so the platform can handle each type differently.
- Added a confidence check that retries uncertain results on long videos, and stored each label with its confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that tells users when their video finishes processing or fails, instead of leaving them waiting on an idle screen.

IDX Exchange | Data Science Intern | Remote **Jun 2026 - Aug 2026**

- Contributed to a California home-price machine learning model trained on **71K** single-family sales; the final model's typical estimate was off by **8.6%** (R^2 of **0.88**).
- Traced an unusable first result to a data leak, fixed it, and brought accuracy back to a usable range.
- Measured accuracy by price tier and across **161** school districts, showing estimates were least reliable for luxury homes (**13%** error vs. **6.5%** mid-priced) and resort markets.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Apify, Factory Droid* GitHub

- Co-designed a transfer planner for California community college students that shows which classes count toward their major using official ASSIST data, citing a source for every answer. Shaped the idea, features, and vision.
- Kept it grounded in real data: credits are matched by code against official records, and campus opportunities are scraped from real pages (Apify, run by a Droid agent), with source links.

SightLine | *Python, PyTorch, OpenCV* GitHub

- ML team member on a mobile app that turns a day of Meta Ray-Ban footage into a **60**-second recap video.

Movie Genre Classification | *Python, PyTorch, scikit-learn, XGBoost, LightGBM* GitHub

- Built a multi-label NLP genre predictor for **42K+** films; chose Macro F1 so rare genres counted equally, raising it from 0.48 to **0.66**.
- Found a simple model beat boosted trees while training about **12x** faster, a speed vs. accuracy tradeoff.

Leadership & Involvement

UC Berkeley CDSS | Dean's Cabinet Transfer Representative | Berkeley, CA **Sep 2026 - Present**

- Advise CDSS leadership on the student experience for CDSS majors.

UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA **Aug 2026 - Present**

- Give resume feedback in a careers seminar, and run exam reviews and practice worksheets for CS and data science classes for transfer students.

UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.